

Cisco Device Management

1. Console Port Security

Console Port Security - login

- By default, no password is needed to access the CLI of a Cisco IOS device via the console port.
- You can configure a password on the *console line*.
 - A user will have to enter a password to access the CLI via the console port.

```
R1(config)#line console 0
R1(config-line)#password ccna
R1(config-line)#login
R1(config-line)#end
R1#exit

R1 con0 is now available
Press RETURN to get started.

User Access Verification
Password:
R1>
```

There is only a single console *line*, so the number is always 0.

Configure the console line's password.

Tell the device to require a user to enter the configured password to access the CLI via the console port.

The password isn't displayed as you type it.

- **Console port** → Used for local/physical CLI access to a router or switch.
- By default, **no password** is required for console access.
- Only one console line exists → **line console 0**

Console Password

```
line console 0
password ccna
login
```

- **password** → Sets console password.
- **login** → Makes the device ask for the configured password.

Important: Password characters are not displayed while typing.

2. login vs login local

login

line console 0
password ccna
login

→ Uses the **password configured on the line**.

login local

The screenshot shows a Cisco IOS configuration session for a device named R1. The configuration steps are as follows:

```
R1(config)#username jeremy secret ccnp
R1(config)#line console 0
R1(config-line)#login local
R1(config-line)#end
R1#exit
```

Annotations in the screenshot:

- A blue box highlights the command `R1(config-line)#login local`, with a blue arrow pointing to a yellow text box that says: "Tell the device to require a user to login using one of the configured usernames on the device."
- A yellow box highlights the configuration block for `line con 0`, containing the following commands:

```
line con 0
exec-timeout 3 30
password ccna
logging synchronous
login local
```

 - A purple box highlights the `exec-timeout 3 30` command, with a purple arrow pointing to a pink text box that says: "Log the user out after 3 minutes and 30 seconds of inactivity."

Below the configuration, the output shows the console line is now available and prompts for user access verification:

```
R1 con0 is now available
Press RETURN to get started.
User Access Verification
Username: jeremy
Password: ****
R1>
```

username jeremy secret ccnp

line console 0
login local

→ Uses a **local username + password** created on the device.

Example:

Username: jeremy
Password: ****

Difference

- `login` → Line password
 - `login local` → Local username database
-

3. VTY Lines

VTY = Virtual Teletype

- VTY lines are used for **remote CLI access**.
- SSH and Telnet connections use VTY lines.
- Example:

```
line vty 0 4
```

→ Configures VTY lines 0–4.

Remember

Console → Local physical access

VTY → Remote access

SSH = how you connect securely

VTY = virtual line through which the remote CLI session enters

4. Remote Access

Remote ≠ Wireless

Remote access means accessing a device **without being physically connected through its console port**.

It can be:

- Wired network
- Wi-Fi
- WAN/Internet

Example:

Laptop → Network → Switch → VTY → CLI

Layer 2 Switch – Management IP



Layer 2 Switch – Management IP

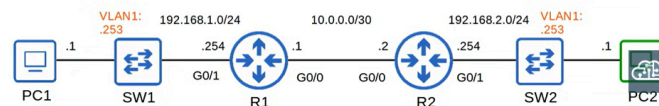
- Layer 2 switches don't perform packet routing and don't build a routing table. They aren't IP routing aware.
- However, you can assign an IP address to an SVI to allow remote connections to the CLI of the switch (using Telnet or SSH).

```
SW1(config)#interface vlan1
SW1(config-if)#ip address 192.168.1.253 255.255.255.0
SW1(config-if)#no shutdown
SW1(config-if)#exit
```

Configure the IP address on the SVI in the same way as on a multilayer switch. Enable the interface if necessary.

```
SW1(config)#ip default-gateway 192.168.1.254
```

Configure the switch's default gateway. In this case, PC2 isn't in the same LAN as SW1. If SW1 doesn't have a default gateway, it can't communicate with PC2.



- A Layer 2 switch normally:
 - Uses **MAC addresses**
 - Performs Layer 2 switching
 - Does NOT perform IP routing
 - Does NOT build a routing table
 - However, a Layer 2 switch can have an IP address for **management**.
-

5. SVI

SVI = Switch Virtual Interface

Example:

```
interface vlan 1
ip address 192.168.1.253 255.255.255.0
no shutdown
```

- SVI provides a **management IP** to the switch.

- This allows:
 - Ping
 - SSH
 - Telnet
 - Other management traffic

Important

Management IP ≠ Routing capability

Giving an IP to a Layer 2 switch does **not** make it a router.

6. Default Gateway on Layer 2 Switch

Example:

```
ip default-gateway 192.168.1.254
```

- Used for the switch's **own IP/management traffic** when the destination is in another network.
- Usually points to the router's IP.

Example:

SW1 = 192.168.1.253

R1 = 192.168.1.254

PC2 = 192.168.2.1

If SW1 needs to communicate with PC2:

SW1 → R1 → R2 → PC2

The default gateway tells SW1:

"For destinations outside my subnet, send the traffic to R1."

7. Important Doubt: Does This Make SW1 a Router?

No.

A Layer 2 switch can send **its own management IP traffic** through a default gateway.

But it cannot route normal user traffic between networks.

Layer 2 Switch

Management traffic:

SW1 → Router → Other Network

✓

User traffic routing:

PC1 → SW1 → Different Network

✗

Layer 3 Switch

ip routing

A Layer 3 switch can actually route traffic between different networks/VLANs.

8. Key Difference

Layer 2 Switch + SVI

→ "You can communicate WITH me."

Layer 3 Switch + IP Routing

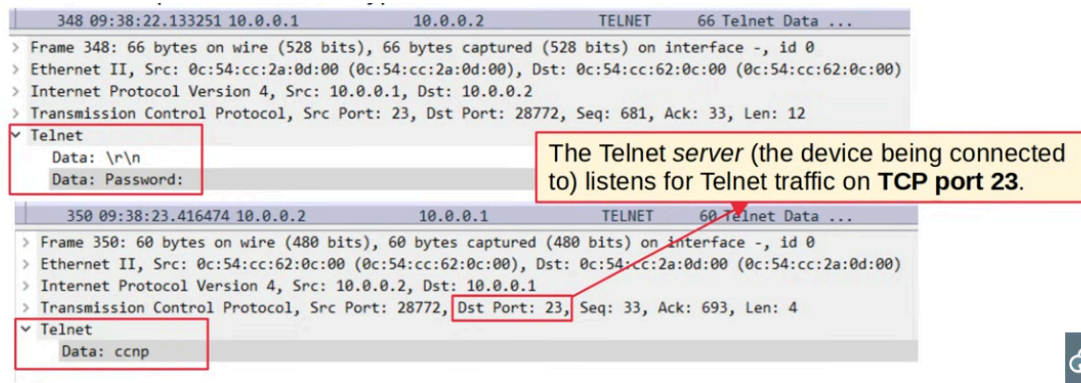
→ "I can route traffic THROUGH me."

One-line revision

SVI → Management IP | Default Gateway → Reach other networks | VTY → Remote CLI |

SSH → Secure remote access | Console → Local access

Telnet



- **Telnet (Telecommunication Network)** → Protocol used for **remote CLI access** to a device.
- Developed in **1969**.
- Uses **TCP port 23**.
- Telnet sends data in **plain text** → **no encryption**.
- Passwords and commands can be captured using tools like Wireshark.
- **SSH replaced Telnet** for secure remote management.

Terminal Client vs Terminal Server

Terminal Client

- The device/application that **initiates** the Telnet connection.
- Example: Your **PC** running a Telnet command.

PC = Telnet Client

Terminal Server

- The device that **receives/accepts** the Telnet connection.
- Example: A **Cisco router/switch** configured to allow Telnet.

Router/Switch = Telnet Server

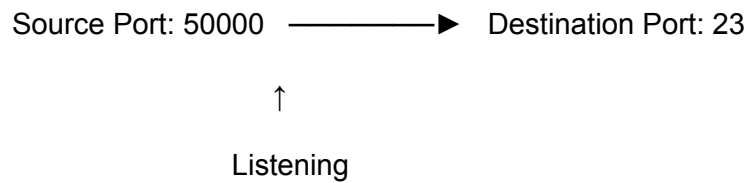
Who listens on TCP Port 23?

The Telnet Server listens on TCP port 23. ✓

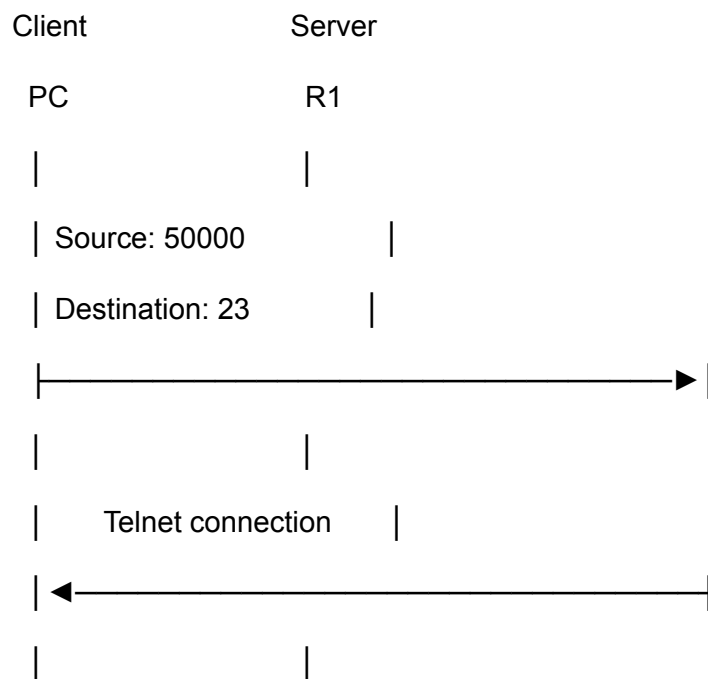
The client uses a **temporary/ephemeral source port**.

Example:

PC (Telnet Client) Router (Telnet Server)



After the connection:



Easy memory

Server listens. Client connects.

So:

Telnet Server → listens on TCP 23

Telnet Client → connects to TCP 23

Telnet vs SSH

Telnet	SSH
TCP 23	TCP 22
No encryption	Encrypted
Plain text	Secure
Less secure	More secure
Older	Recommended

Important

Remote CLI



Telnet / SSH



VTY lines



Device CLI

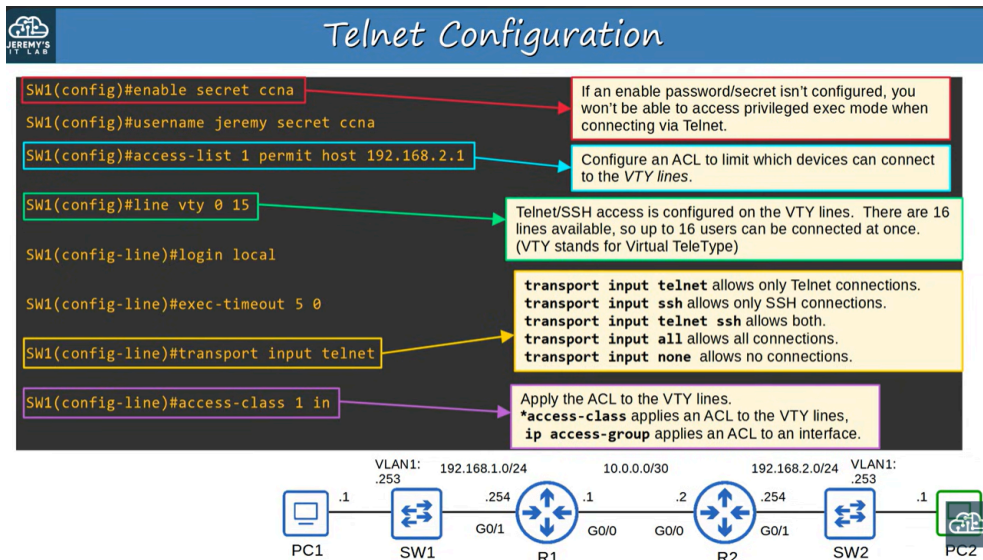
Remember:

Telnet = Remote + TCP 23 + Plain text

SSH = Remote + TCP 22 + Encrypted

Image placement: Paste the Telnet image directly after this section, especially below the **TCP port 23 / plain-text** points.

Telnet Configuration



Basic Configuration

```
enable secret ccna
username jeremy secret ccna
```

```
line vty 0 15
login local
exec-timeout 5 0
transport input telnet
```

- **enable secret** → Password required to enter **Privileged EXEC (#)** mode.
- **username ... secret** → Creates a local username/password for login.
- **line vty 0 15** → Configures **16 VTY lines** → up to 16 simultaneous remote sessions.
- **login local** → Uses the local username database.
- **exec-timeout 5 0** → Disconnects inactive session after **5 minutes**.
- **transport input telnet** → Allows only Telnet connections.

transport input

```
telnet → Telnet only
ssh → SSH only
telnet ssh → Telnet + SSH
all → All supported connections
none → No remote connections
```

VTY Access Control with ACL

1. Create ACL

```
access-list 1 permit host 192.168.2.1
```

→ Allows only **192.168.2.1** to access the VTY lines.

2. Apply ACL to VTY

```
line vty 0 15  
access-class 1 in
```

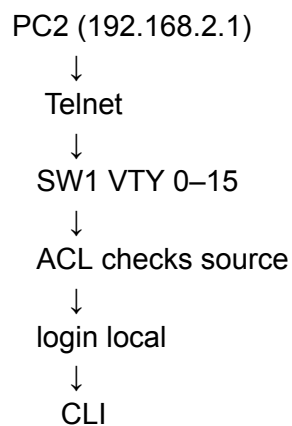
→ Applies the ACL to **incoming VTY connections**.

Important Difference

access-class → ACL applied to VTY lines

ip access-group → ACL applied to a physical/L3 interface

Complete Flow



Remember:

VTY = remote access | **transport input** = allowed protocol | **login local** = username authentication | **access-class** = restrict who can access VTY

Telnet Test with ACL



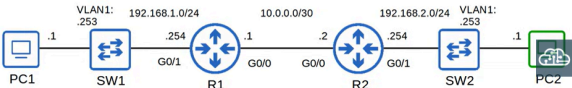
Telnet Configuration

```
R2#ping 192.168.1.253
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.253, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 10/11/16 ms

R2#telnet 192.168.1.253
Trying 192.168.1.253 ...
% Connection refused by remote host
```

```
C:\Users\User>telnet 192.168.1.253
Connecting To 192.168.0.1...
User Access Verification
Username: jeremy
Password:
SW1>
```

```
line vty 0 4
access-class 1 in
exec-timeout 5 0
login local
transport input telnet
line vty 5 15
access-class 1 in
exec-timeout 5 0
login local
transport input telnet
```



Test 1: Ping

R2:

```
R2# ping 192.168.1.253
```

→ **100% successful**

This means R2 can reach SW1's management IP at Layer 3.

Test 2: Telnet from R2

```
R2# telnet 192.168.1.253
```

% Connection refused by remote host

Why?

SW1 has:

```
access-list 1 permit host 192.168.2.1
```

```
line vty 0 15
```

access-class 1 in

The ACL allows **only 192.168.2.1** to access SW1's VTY lines.

R2's source IP is **192.168.2.2**, so it is denied.

PC2 Telnet

PC2 has:

192.168.2.1

So:

PC2 (192.168.2.1)

↓

Telnet

↓

SW1 (192.168.1.253)

↓

ACL → PERMIT

↓

VTY

↓

Username + Password

↓

SW1>

Therefore, PC2 successfully gets the SW1 CLI.

Important Concept

Ping successful ≠ Telnet successful

Ping

→ Tests network reachability

→ ICMP

Telnet

→ Tests remote CLI access

→ TCP port 23

→ Also affected by VTY configuration and ACL

So R2 can **reach** SW1 but cannot **Telnet** into it because the VTY ACL blocks R2.

Easy memory

Ping asks: "Can I reach the device?"

Telnet asks: "Can I remotely log in to the device?"

ACL can allow one device to Telnet while blocking another device, even when both can ping the switch.

SSH (Secure Shell)

- **SSH = Secure Shell**
- Developed in **1995** to replace insecure remote-access protocols like Telnet.
- **SSHv2** → major revision of SSHv1, released in 2006.
- Provides:
 - **Encryption** → protects data from being read.
 - **Authentication** → verifies the user/device.

- Uses **TCP port 22**.
- SSH is used for **secure remote CLI access**.

SSH vs Telnet

Telnet → TCP 23 → Plain text → Not secure

SSH → TCP 22 → Encrypted → Secure

What is a Shell?

A **shell** is a program/interface that allows a user to interact with the operating system using commands.

Example:

User → Shell/CLI → Operating System

In networking, SSH gives you a **remote shell/CLI session** on the router or switch.

SSH Versions

- **SSHv1** → Older, less secure.
- **SSHv2** → Newer and recommended.
- **Version 1.99** → Device supports both SSHv1 and SSHv2.

Wireshark

SSH (Secure Shell)

- SSH (Secure Shell) was developed in 1995 to replace less secure protocols like Telnet.

In computing, a **shell** is a computer program which exposes an operating system's services to a human user or other program. In general, operating system shells use either a **command-line interface** (CLI) or **graphical user interface** (GUI), depending on a computer's role and particular operation. It is named a shell because it is the outermost layer around the operating system.^{[1] [2]}
- SSHv2, a major revision of SSHv1, was released in 2006.
- If a device supports both version 1 and version 2, it is said to run 'version 1.99'.
- Provides security features such as data encryption and authentication.

SSH traffic appears as:

SSHv2

TCP Source Port: 22

Encrypted Packet

The actual commands/passwords are **not visible as plain text** because SSH encrypts the communication.

Easy Memory

SSH = Secure + Remote CLI + TCP 22 + Encryption + Authentication

Complete Remote Access Flow

SSH Client



TCP 22



SSH Server



VTY Line



Authentication



CLI

SSH Configuration – Check SSH Support

```
SW1#show version
Cisco IOS Software, vios_12 Software (vios_12-ADVENTERPRISEK9-M), Version 15.2(4.0.55)E, TEST
ENGINEERING ESTG_WEEKLY BUILD, synced to END_OF_FLO_ISP
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2015 by Cisco Systems, Inc.
Compiled Tue 28-Jul-15 18:52 by sasyamal

SW1#show ip ssh
SSH Disabled - version 1.99
%Please create RSA keys to enable SSH (and of atleast 768 bits for SSH v2).
Authentication methods:publickey,keyboard-interactive,password
Authentication Publickey Algorithms:x509v3-ssh-rsa,ssh-rsa
Hostkey Algorithms:x509v3-ssh-rsa,ssh-rsa
Encryption Algorithms:aes128-ctr,aes192-ctr,aes256-ctr,aes128-cbc,3des-cbc,aes192-cbc,aes256-cbc
MAC Algorithms:hmac-sha1,hmac-sha1-96
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded): NONE
```

Check IOS Version

show version

→ Displays the Cisco IOS version and image name.

Check SSH Support

show ip ssh

→ Displays SSH configuration/status.

Example:

SSH Disabled - version 1.99

Version 1.99 → Device supports both SSHv1 and SSHv2.

RSA Keys Required

If you see:

Please create RSA keys to enable SSH

→ Generate RSA keys before SSH can be enabled.

crypto key generate rsa

- RSA keys are used for **SSH encryption/authentication**.
 - For SSHv2, the image requires an appropriate RSA key size.
-

K9 vs NPE IOS

K9

- IOS image containing **cryptographic/encryption support**.
- Supports features such as **SSH**.

NPE

NPE = No Payload Encryption

- IOS image without cryptographic features.
- Does **not support SSH**.

Easy Memory

K9 → Crypto support → SSH ✓

NPE → No encryption → SSH ✗

Important: Check the IOS image name for **K9** when verifying cryptographic/SSH support.

SSH Configuration – RSA Keys



SSH Configuration: RSA Keys

- To enable and use SSH, you must generate an RSA public and private key pair.
- The keys are used for data encryption/decryption, authentication, etc.

```
SW1(config)#ip domain name jeremysitlab.com
SW1(config)#crypto key generate rsa
The name for the keys will be: SW1.jeremysitlab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...
[OK] (elapsed time was 1 seconds)
SW1(config)#
*Feb 21 04:22:35.778: %SSH-5-ENABLED: SSH 1.99 has been enabled
SW1(config)#do show ip ssh
SSH Enabled - version 1.99
Authentication methods:publickey,keyboard-interactive,password
Encryption Algorithms:aes128-ctr,aes192-ctr,aes256-ctr,aes128-cbc,3des-cbc,aes192-cbc,aes256-cbc
MAC Algorithms:hmac-sha1,hmac-sha1-96
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded): SW1.jeremysitlab.com
[output omitted]
```

The **FQDN** of the device is used to name the RSA keys.
FQDN = Fully Qualified Domain Name (host name + domain name)

Generate the RSA keys.
crypto key generate rsa modulus Length
is an alternate method.
*length must be 768 bits or greater for SSHv2

- SSH requires an **RSA public/private key pair**.
- RSA keys help provide **encryption and authentication**.

1. Configure Domain Name

ip domain name jeremysitlab.com

- Used with the hostname to create the device's **FQDN**.
- **FQDN = Fully Qualified Domain Name**
- Example:

SW1 + jeremysitlab.com

→ SW1.jeremysitlab.com

2. Generate RSA Keys

crypto key generate rsa

or:

crypto key generate rsa modulus 2048

- Generates the RSA key pair.
- **Key size** → measured in bits.
- Larger key size generally provides stronger security but may take longer to generate.
- **SSH v2 requires at least 768-bit RSA keys** in the context shown.

Example:

How many bits in the modulus [512]: 2048

→ Generates a **2048-bit RSA key**.

3. Verify SSH

show ip ssh

Example:

SSH Enabled - version 1.99

→ SSH is now enabled.

The output also shows:

IOS Keys in SECSH format:

SW1.jeremysitlab.com

→ The RSA keys are associated with the device's FQDN.

Key Flow

Hostname + Domain Name

↓

FQDN

↓

crypto key generate rsa



RSA Public + Private Keys



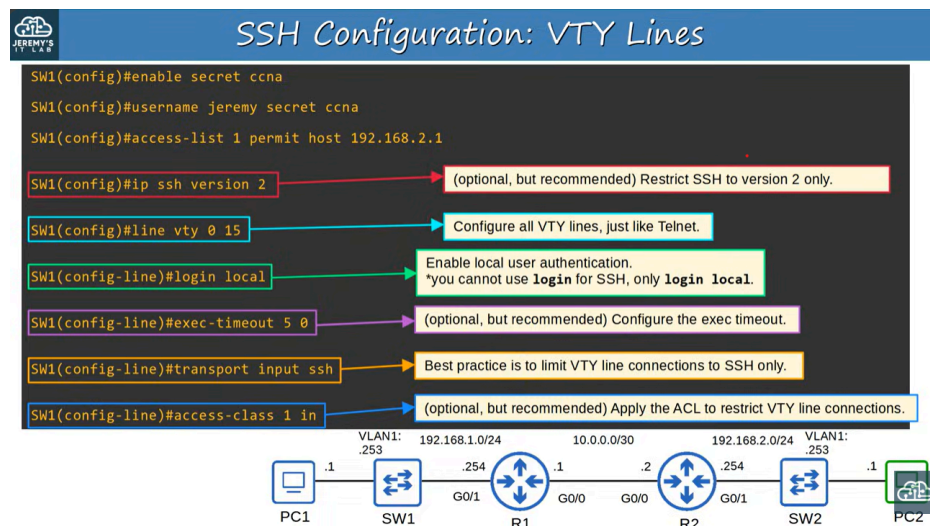
SSH enabled

Remember

Domain name → FQDN → RSA keys → SSH

crypto key generate rsa = creates the keys required for SSH.

SSH Configuration – VTY Lines



enable secret ccna

username jeremy secret ccna

access-list 1 permit host 192.168.2.1

ip ssh version 2

line vty 0 15

login local

exec-timeout 5 0

transport input ssh

access-class 1 in

Commands

- **enable secret** → Password for **Privileged EXEC (#) mode**.
- **username ... secret** → Creates local username/password.
- **ip ssh version 2** → Allows **SSHv2 only**.
- **line vty 0 15** → Configures all **16 VTY lines**.
- **login local** → Uses the local username database.
⚠ For SSH, use **login local**, not just **login**.
- **exec-timeout 5 0** → Ends inactive session after **5 minutes**.
- **transport input ssh** → Allows **SSH only**, not Telnet.
- **access-class 1 in** → Applies ACL 1 to incoming VTY connections.

SSH VTY Flow

SSH Client



TCP 22



SW1 Management IP



ACL check



VTY 0–15



login local



Username + Password



CLI

Key Point

Telnet:

transport input telnet

SSH:

transport input ssh

Best practice: Use **SSHv2 + login local + transport input ssh + VTY ACL** for secure remote management.

SSH Configuration – Final Summary

 *SSH Configuration*

- 1) **Configure host name**
- 2) Configure DNS domain name
- 3) Generate RSA key pair
- 4) Configure enable PW, username/PW
- 5) Enable SSHv2 (only)
- 6) Configure VTY lines

```
Router(config)#crypto key generate rsa
% Please define a hostname other than Router.
Router(config)#hostname R2
R2(config)#crypto key generate rsa
% Please define a domain-name first.
R2(config)#ip domain name jeremysitlab.com
R2(config)#crypto key generate rsa
The name for the keys will be: R2.jeremysitlab.com
[output omitted]
```

Connect: **ssh -l username ip-address** OR **ssh username@ip-address**

SSH Configuration Order

1. Configure hostname

hostname R2

2. Configure DNS domain name

ip domain name jeremysitlab.com

3. Generate RSA key pair

crypto key generate rsa

4. Configure passwords

enable secret ccna

username jeremy secret ccna

5. Enable SSHv2

ip ssh version 2

6. Configure VTY lines

line vty 0 15

login local

transport input ssh

exec-timeout 5 0

Why hostname + domain name?

hostname + domain name



FQDN



Used to name RSA keys

Example:

R2.jeremysitlab.com

 **Hostname and domain name must be configured before generating RSA keys.**

SSH Connection

From a client:

```
ssh -l username ip-address
```

or:

```
ssh username@ip-address
```

Example:

```
ssh -l jeremy 192.168.1.253
```

```
ssh jeremy@192.168.1.253
```

→ Connects remotely to the device using **SSH**.

Important Command Summary

 *Command Summary*

```
SW1# show version
SW1# show ip ssh
SW1(config)# ip default-gateway ip-address
SW1(config)# line con 0
SW1(config)# line vty 0 15
SW1(config)# crypto key generate rsa
SW1(config)# ip ssh version 2
SW1(config-line)# login [local]
SW1(config-line)# transport input [protocols | all | none]
SW1(config-line)# exec-timeout minutes seconds
SW1(config-line)# access-class acl in

> telnet ip-address
> ssh -l username ip-address
> ssh username@ip-address
```



Verification

show version

→ Check IOS version/image.

show ip ssh

→ Check SSH status/version.

Layer 2 Switch Management

ip default-gateway ip-address

→ Default gateway for a Layer 2 switch's **management traffic**.

Console

```
line con 0
```

→ Configure the console line.

Remote Access

```
line vty 0 15
```

→ Configure all **16 VTY lines**.

RSA

```
crypto key generate rsa
```

→ Generate RSA keys required for SSH.

SSH Version

```
ip ssh version 2
```

→ Use SSHv2 only.

Authentication

```
login local
```

→ Authenticate using locally configured usernames.

Remote Protocol

```
transport input ssh
```

→ Allow SSH connections.

transport input telnet

→ Allow Telnet connections.

transport input telnet ssh

→ Allow both.

transport input none

→ Allow no remote connections.

Timeout

exec-timeout minutes seconds

→ Disconnect inactive sessions after the specified time.

Example:

exec-timeout 5 0

→ Disconnect after **5 minutes of inactivity**.

VTY ACL

access-class acl-number in

→ Apply an ACL to **incoming VTY connections** and restrict who can remotely access the device.

SSH Complete Flow

Hostname



Domain Name



FQDN



RSA Key Pair



Username + Password



SSHv2



VTY Lines



SSH Client



TCP 22



Remote CLI

Final Memory

SSH = Remote CLI + TCP 22 + Encryption

Telnet = Remote CLI + TCP 23 + Plain text

VTY = Virtual lines used for remote access

RSA keys = Required to enable SSH

login local = Use local username/password

transport input ssh = Allow SSH

access-class = Control who can access VTY

ip default-gateway = Allows a Layer 2 switch to send management traffic to another network